John Munger

AI Platform Engineer | AI Runtime Systems Brief

AI Runtime Infrastructure • Agent Orchestration • Retrieval Systems • Runtime Governance

Overview

I design AI platform systems that enable Large Language Models to operate safely within production environments through governed execution, retrieval grounding, runtime observability, and deterministic execution boundaries. Rather than treating LLMs as autonomous systems, my work focuses on building runtime infrastructure that constrains, validates, and explains AI behavior through explicit execution policies, kernel authority, and traceable system evidence. My engineering philosophy emphasizes building reliable AI platforms that remain observable, explainable, and maintainable as capabilities evolve.

AI Control Plane Runtime

Designed and implemented a modular AI runtime separating transport, orchestration, policy, kernel, and tool registry responsibilities into independent architectural layers. The runtime establishes deterministic execution boundaries where all tool invocation, validation, enforcement, and user-visible output flow through a centralized kernel.

Core Capabilities
• Policy-governed execution boundaries
• Kernel-enforced validation and schema contracts
• Runtime observability through trace-driven execution
• Structured tool registry and agent orchestration
• Extensible architecture supporting retrieval and tool-calling workflows
• Multi-model routing balancing quality, latency, and operational cost

Runtime Observability

Developed a trace-driven execution model capturing planner decisions, tool invocation, validation, policy enforcement, execution results, and runtime outputs. This provides explainable evidence for debugging, evaluation, and operational confidence.

Retrieval Infrastructure

Built retrieval-grounded developer tooling with document ingestion, parsing, normalization, intelligent chunking, embedding generation, vector similarity search, context construction, and citation-backed responses using PostgreSQL pgvector and Voyage embeddings.

Agent Workflows

Designed bounded AI workflows for automated code review, documentation generation, repository analysis, controlled CI/CD automation, execution validation, rollback safeguards, and evaluation workflows.

Engineering Principles

Runtime Governance • Execution Authority • Validation-First Design • Retrieval-Grounded

Reasoning • Observability by Default • Explainable AI Systems • Cost-Aware Model Routing • Safe Operational Boundaries

Technical Stack

TypeScript, Node.js, OpenAI, Anthropic, Voyage AI, RAG Pipelines, Agent Workflows, PostgreSQL, pgvector, GitHub Actions, CI/CD Automation, Zod Runtime Validation, Distributed Systems

Focus Areas

AI Platform Engineering • AI Runtime Infrastructure • Agent Orchestration • Retrieval-Augmented Generation • Runtime Governance • LLM Systems • Observability & Evaluation • Production AI Architecture